

Creation Date 22-Jul-2024

Revision Date 23-Jul-2024

Revision Number 1

**Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING****1.1. Product identifier**

Product Description: 50% v/v Ethylene Glycol Solution  
Cat No. : TS/0607/17: TS/0607/99

**1.2. Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

**1.3. Details of the supplier of the safety data sheet****Company**

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road, Loughborough,  
Leicestershire LE11 5RG, United Kingdom

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

**E-mail address** begel.sdsdesk@thermofisher.com

**1.4. Emergency telephone number**

Tel: 01509 231166  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001-703-527-3887

For customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

**Section 2: HAZARDS IDENTIFICATION****2.1. Classification of the substance or mixture**

CLP Classification - Regulation (EC) No 1272/2008

# SAFETY DATA SHEET

50% v/v Ethylene Glycol Solution

Revision Date 23-Jul-2024

## Physical hazards

Based on available data, the classification criteria are not met

## Health hazards

Specific target organ toxicity - (repeated exposure)

Category 2 (H373)

## Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Warning

## Hazard Statements

H373 - May cause damage to organs through prolonged or repeated exposure

## Precautionary Statements

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P314 - Get medical advice/attention if you feel unwell

## 2.3. Other hazards

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## **Section 3: Composition/information on ingredients**

### 3.2. Mixtures

| Component       | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008 |
|-----------------|-----------|-------------------|----------|---------------------------------------------------|
| Ethylene glycol | 107-21-1  | EEC No. 203-473-3 | 50 - 60  | Acute Tox. 4 (H302)<br>STOT RE 2 (H373)           |
| Water           | 7732-18-5 | 231-791-2         | 40 - 50  | -                                                 |

| Components      | Reach Registration Number |
|-----------------|---------------------------|
| Ethylene glycol | 01-2119456816-28          |

# SAFETY DATA SHEET

50% v/v Ethylene Glycol Solution

Revision Date 23-Jul-2024

Full text of Hazard Statements: see section 16

## Section 4: First aid measures

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                                                           |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### 4.3. Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically. Symptoms may be delayed.

## Section 5: Firefighting measures

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

#### **Hazardous Combustion Products**

None under normal use conditions.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required.

# SAFETY DATA SHEET

50% v/v Ethylene Glycol Solution

Revision Date 23-Jul-2024

## 6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

## 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7: Handling and storage

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 10

**Switzerland - Storage of hazardous substances**

Storage class - SC 10/12  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## Section 8: Exposure controls/personal protection

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component       | European Union                                                                                            | The United Kingdom                                                                             | France                                                                                                   | Belgium                                                                           | Spain                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Ethylene glycol | TWA: 20 ppm (8h)<br>TWA: 52 mg/m <sup>3</sup> (8h)<br>STEL: 40 ppm (15min)<br>STEL: 104 mg/m <sup>3</sup> | STEL: 40 ppm 15 min<br>STEL: 104 mg/m <sup>3</sup> 15 min<br>STEL: 30 mg/m <sup>3</sup> 15 min | TWA / VME: 20 ppm (8 heures). indicative limit<br>TWA / VME: 52 mg/m <sup>3</sup> (8 heures). indicative | TWA: 20 ppm 8 uren<br>TWA: 52 mg/m <sup>3</sup> 8 uren<br>STEL: 40 ppm 15 minuten | STEL / VLA-EC: 40 ppm (15 minutos).<br>STEL / VLA-EC: 104 mg/m <sup>3</sup> (15 minutos). |

# SAFETY DATA SHEET

50% v/v Ethylene Glycol Solution

Revision Date 23-Jul-2024

|  |                 |                                                                                              |                                                                                                                       |                                                   |                                                                                              |
|--|-----------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------|
|  | (15min)<br>Skin | TWA: 10 mg/m <sup>3</sup> 8 hr<br>TWA: 20 ppm 8 hr<br>TWA: 52 mg/m <sup>3</sup> 8 hr<br>Skin | limit<br>STEL / VLCT: 40 ppm.<br>indicative limit<br>STEL / VLCT: 104<br>mg/m <sup>3</sup> . indicative limit<br>Peau | STEL: 104 mg/m <sup>3</sup> 15<br>minuten<br>Huid | TWA / VLA-ED: 20 ppm<br>(8 horas)<br>TWA / VLA-ED: 52<br>mg/m <sup>3</sup> (8 horas)<br>Piel |
|--|-----------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------|

| Component       | Italy                                                                                                                                                                                                              | Germany                                                                                                                                                                                                                                                                                                                                                                                              | Portugal                                                                                                                                                                      | The Netherlands                                                                                                                                         | Finland                                                                                                                                                     |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethylene glycol | TWA: 20 ppm 8 ore.<br>Time Weighted Average<br>TWA: 52 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>STEL: 40 ppm 15<br>minuti. Short-term<br>STEL: 104 mg/m <sup>3</sup> 15<br>minuti. Short-term<br>Pelle | TWA: 10 ppm (8<br>Stunden). AGW -<br>exposure factor 2<br>TWA: 26 mg/m <sup>3</sup> (8<br>Stunden). AGW -<br>exposure factor 2<br>TWA: 10 ppm (8<br>Stunden). MAK can<br>occur as vapor and<br>aerosol at the same<br>time<br>TWA: 26 mg/m <sup>3</sup> (8<br>Stunden). MAK can<br>occur as vapor and<br>aerosol at the same<br>time<br>Höhepunkt: 20 ppm<br>Höhepunkt: 52 mg/m <sup>3</sup><br>Haut | STEL: 40 ppm 15<br>minutos<br>STEL: 104 mg/m <sup>3</sup> 15<br>minutos<br>Ceiling: 100 mg/m <sup>3</sup><br>TWA: 20 ppm 8 horas<br>TWA: 52 mg/m <sup>3</sup> 8 horas<br>Pele | huid<br>STEL: 40 ppm 15<br>minuten<br>STEL: 104 mg/m <sup>3</sup> 15<br>minuten<br>TWA: 52 mg/m <sup>3</sup> 8 uren<br>TWA: 10 mg/m <sup>3</sup> 8 uren | TWA: 20 ppm 8 tunteina<br>TWA: 50 mg/m <sup>3</sup> 8<br>tunteina<br>STEL: 40 ppm 15<br>minuutteina<br>STEL: 100 mg/m <sup>3</sup> 15<br>minuutteina<br>Iho |

| Component       | Austria                                                                                                                                                              | Denmark                                                                                                                                                                                                                        | Switzerland                                                                                                                                               | Poland                                                                                | Norway                                                                                                                                                                                                                                                                                                                                                           |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethylene glycol | Haut<br>MAK-KZGW: 20 ppm 15<br>Minuten<br>MAK-KZGW: 52 mg/m <sup>3</sup><br>15 Minuten<br>MAK-TMW: 10 ppm 8<br>Stunden<br>MAK-TMW: 26 mg/m <sup>3</sup> 8<br>Stunden | TWA: 10 ppm 8 timer<br>TWA: 26 mg/m <sup>3</sup> 8 timer<br>TWA: 10 mg/m <sup>3</sup> 8 timer<br>STEL: 104 mg/m <sup>3</sup> 15<br>minutter<br>STEL: 40 ppm 15<br>minutter<br>STEL: 20 mg/m <sup>3</sup> 15<br>minutter<br>Hud | Haut/Peau<br>STEL: 20 ppm 15<br>Minuten<br>STEL: 52 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 10 ppm 8<br>Stunden<br>TWA: 26 mg/m <sup>3</sup> 8<br>Stunden | STEL: 50 mg/m <sup>3</sup> 15<br>minutach<br>TWA: 15 mg/m <sup>3</sup> 8<br>godzinach | TWA: 20 ppm 8 timer<br>TWA: 52 mg/m <sup>3</sup> 8 timer<br>STEL: 104 mg/m <sup>3</sup> 15<br>minutter. total sum of<br>gas and particulate<br>matter (aerosol) of the<br>substance;value from<br>the regulation<br>STEL: 40 ppm 15<br>minutter. total sum of<br>gas and particulate<br>matter (aerosol) of the<br>substance;value from<br>the regulation<br>Hud |

| Component       | Bulgaria                                                                                                   | Croatia                                                                                                                                                                     | Ireland                                                                                                                      | Cyprus                                                                                                                                | Czech Republic                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Ethylene glycol | TWA: 52 mg/m <sup>3</sup><br>TWA: 20 ppm<br>STEL : 40 ppm<br>STEL : 104 mg/m <sup>3</sup><br>Skin notation | kože<br>TWA-GVI: 20 ppm 8<br>satima.<br>TWA-GVI: 52 mg/m <sup>3</sup> 8<br>satima.<br>STEL-KGVI: 40 ppm 15<br>minutama.<br>STEL-KGVI: 104 mg/m <sup>3</sup><br>15 minutama. | TWA: 20 ppm 8 hr.<br>TWA: 52 mg/m <sup>3</sup> 8 hr.<br>STEL: 40 ppm 15 min<br>STEL: 104 mg/m <sup>3</sup> 15<br>min<br>Skin | Skin-potential for<br>cutaneous absorption<br>STEL: 40 ppm<br>STEL: 104 mg/m <sup>3</sup><br>TWA: 20 ppm<br>TWA: 52 mg/m <sup>3</sup> | TWA: 50 mg/m <sup>3</sup> 8<br>hodinách.<br>Potential for cutaneous<br>absorption<br>Ceiling: 100 mg/m <sup>3</sup> |

| Component       | Estonia                                                                                                                                                                                                                                         | Gibraltar                                                                                                                           | Greece                                                                                   | Hungary                                                                                                                                                                                                       | Iceland                                                                                                                                                                                                                  |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethylene glycol | Nahk<br>TWA: 20 ppm 8<br>tundides. total<br>concentration of aerosol<br>and vapor<br>TWA: 52 mg/m <sup>3</sup> 8<br>tundides. total<br>concentration of aerosol<br>and vapor<br>STEL: 40 ppm 15<br>minutites. total<br>concentration of aerosol | Skin notation<br>TWA: 20 ppm 8 hr<br>TWA: 52 mg/m <sup>3</sup> 8 hr<br>STEL: 40 ppm 15 min<br>STEL: 104 mg/m <sup>3</sup> 15<br>min | STEL: 50 ppm<br>STEL: 125 mg/m <sup>3</sup><br>TWA: 50 ppm<br>TWA: 125 mg/m <sup>3</sup> | STEL: 40 ppm 15<br>percekben. CK<br>STEL: 104 mg/m <sup>3</sup> 15<br>percekben. CK<br>TWA: 20 ppm 8 órában.<br>AK<br>TWA: 52 mg/m <sup>3</sup> 8<br>órában. AK<br>lehetséges borón<br>keresztüli felszívódás | STEL: 40 ppm<br>STEL: 104 mg/m <sup>3</sup><br>TWA: 10 ppm 8<br>klukkustundum. mist<br>TWA: 26 mg/m <sup>3</sup> 8<br>klukkustundum. mist<br>Skin notation<br>Ceiling: 20 ppm mist<br>Ceiling: 52 mg/m <sup>3</sup> mist |

# SAFETY DATA SHEET

50% v/v Ethylene Glycol Solution

Revision Date 23-Jul-2024

|  |                                                                                                        |  |  |  |  |
|--|--------------------------------------------------------------------------------------------------------|--|--|--|--|
|  | and vapor<br>STEL: 104 mg/m <sup>3</sup> 15<br>minutes. total<br>concentration of aerosol<br>and vapor |  |  |  |  |
|--|--------------------------------------------------------------------------------------------------------|--|--|--|--|

| Component       | Latvia                                                                                                                                | Lithuania                                                                                                                                         | Luxembourg                                                                                                                                                                                             | Malta                                                                                                                                                                   | Romania                                                                                                                                        |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethylene glycol | skin - potential for<br>cutaneous exposure<br>STEL: 40 ppm<br>STEL: 104 mg/m <sup>3</sup><br>TWA: 20 ppm<br>TWA: 52 mg/m <sup>3</sup> | TWA: 10 ppm aerosol<br>and vapor IPRD<br>TWA: 25 mg/m <sup>3</sup> aerosol<br>and vapor IPRD<br>Oda<br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup> | Possibility of significant<br>uptake through the skin<br>TWA: 20 ppm 8<br>Stunden<br>TWA: 52 mg/m <sup>3</sup> 8<br>Stunden<br>STEL: 40 ppm 15<br>Minuten<br>STEL: 104 mg/m <sup>3</sup> 15<br>Minuten | possibility of significant<br>uptake through the skin<br>TWA: 20 ppm<br>TWA: 52 mg/m <sup>3</sup><br>STEL: 40 ppm 15 minuti<br>STEL: 104 mg/m <sup>3</sup> 15<br>minuti | Skin notation<br>TWA: 20 ppm 8 ore<br>TWA: 52 mg/m <sup>3</sup> 8 ore<br>STEL: 40 ppm 15<br>minute<br>STEL: 104 mg/m <sup>3</sup> 15<br>minute |

| Component       | Russia                                                     | Slovak Republic                                                                                                     | Slovenia                                                                                                                                  | Sweden                                                                                                                                                                       | Turkey                                                                                                                                  |
|-----------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Ethylene glycol | TWA: 5 mg/m <sup>3</sup> 2388<br>MAC: 10 mg/m <sup>3</sup> | Ceiling: 104 mg/m <sup>3</sup><br>Potential for cutaneous<br>absorption<br>TWA: 20 ppm<br>TWA: 52 mg/m <sup>3</sup> | TWA: 20 ppm 8 urah<br>TWA: 52 mg/m <sup>3</sup> 8 urah<br>Koža<br>STEL: 40 ppm 15<br>minutah<br>STEL: 104 mg/m <sup>3</sup> 15<br>minutah | Binding STEL: 40 ppm<br>15 minuter<br>Binding STEL: 104<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 10 ppm 8 timmar.<br>NGV<br>TLV: 25 mg/m <sup>3</sup> 8<br>timmar. NGV<br>Hud | Deri<br>TWA: 20 ppm 8 saat<br>TWA: 52 mg/m <sup>3</sup> 8 saat<br>STEL: 40 ppm 15<br>dakika<br>STEL: 104 mg/m <sup>3</sup> 15<br>dakika |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                               | Acute effects local<br>(Dermal) | Acute effects<br>systemic (Dermal) | Chronic effects local<br>(Dermal) | Chronic effects<br>systemic (Dermal) |
|-----------------------------------------|---------------------------------|------------------------------------|-----------------------------------|--------------------------------------|
| Ethylene glycol<br>107-21-1 ( 50 - 60 ) |                                 |                                    |                                   | DNEL = 106mg/kg<br>bw/day            |

| Component                               | Acute effects local<br>(Inhalation) | Acute effects<br>systemic (Inhalation) | Chronic effects local<br>(Inhalation)                      | Chronic effects<br>systemic (Inhalation) |
|-----------------------------------------|-------------------------------------|----------------------------------------|------------------------------------------------------------|------------------------------------------|
| Ethylene glycol<br>107-21-1 ( 50 - 60 ) |                                     |                                        | DNEL = 35mg/m <sup>3</sup><br>DNEL = 33.5mg/m <sup>3</sup> | DNEL = 70mg/m <sup>3</sup>               |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                               | Fresh water                      | Fresh water<br>sediment       | Water Intermittent              | Microorganisms in<br>sewage treatment | Soil (Agriculture)          |
|-----------------------------------------|----------------------------------|-------------------------------|---------------------------------|---------------------------------------|-----------------------------|
| Ethylene glycol<br>107-21-1 ( 50 - 60 ) | PNEC = 10mg/L<br>PNEC = 85.9mg/L | PNEC = 37mg/kg<br>sediment dw | PNEC = 10mg/L<br>PNEC = 130mg/L | PNEC = 199.5mg/L<br>PNEC = 200mg/L    | PNEC = 1.53mg/kg<br>soil dw |

# SAFETY DATA SHEET

50% v/v Ethylene Glycol Solution

Revision Date 23-Jul-2024

|  |  |                                                                  |  |  |                                                            |
|--|--|------------------------------------------------------------------|--|--|------------------------------------------------------------|
|  |  | PNEC = 312mg/kg<br>sediment dw<br>PNEC = 317mg/kg<br>sediment dw |  |  | PNEC = 12.7mg/kg<br>soil dw<br>PNEC = 13.1mg/kg<br>soil dw |
|--|--|------------------------------------------------------------------|--|--|------------------------------------------------------------|

| Component                               | Marine water                    | Marine water sediment                                                                                | Marine water Intermittent | Food chain | Air |
|-----------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------|---------------------------|------------|-----|
| Ethylene glycol<br>107-21-1 ( 50 - 60 ) | PNEC = 1mg/L<br>PNEC = 8.59mg/L | PNEC = 3.7mg/kg<br>sediment dw<br>PNEC = 31.2mg/kg<br>sediment dw<br>PNEC = 31.7mg/kg<br>sediment dw | PNEC = 10mg/L             |            |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Viton (R)      | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

No information available.

## Section 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

# SAFETY DATA SHEET

50% v/v Ethylene Glycol Solution

Revision Date 23-Jul-2024

|                                         |                          |                      |
|-----------------------------------------|--------------------------|----------------------|
| Physical State                          | Liquid                   |                      |
| Appearance                              | Clear                    |                      |
| Odor                                    | Odorless                 |                      |
| Odor Threshold                          | No data available        |                      |
| Melting Point/Range                     | -13 °C / 8.6 °F          | Estimated            |
| Softening Point                         | No data available        |                      |
| Boiling Point/Range                     | 195 °C / 383 °F          | @ 760 mmHg Estimated |
| Flammability (liquid)                   | No data available        |                      |
| Flammability (solid,gas)                | Not applicable           | Liquid               |
| Explosion Limits                        | No data available        |                      |
| Flash Point                             | 111 °C / 231.8 °F        | Method - Estimated   |
| Autoignition Temperature                | 410 °C / 770 °F          | Estimated            |
| Decomposition Temperature               | No data available        |                      |
| pH                                      |                          |                      |
| Viscosity                               | No data available        |                      |
| Water Solubility                        | Soluble                  |                      |
| Solubility in other solvents            | No information available |                      |
| Partition Coefficient (n-octanol/water) |                          |                      |
| Component                               | log Pow                  |                      |
| Ethylene glycol                         | -1.36                    |                      |
| Vapor Pressure                          | .05 mmHg @ 20 °C         |                      |
| Density / Specific Gravity              | 1.05                     | Estimated            |
| Bulk Density                            | Not applicable           | Liquid               |
| Vapor Density                           | 2.1                      | (Air = 1.0)          |
| Particle characteristics                | Not applicable (liquid)  |                      |

## 9.2. Other information

Evaporation Rate No information available

## Section 10: Stability and reactivity

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.  
Hazardous Reactions None under normal processing.

### 10.4. Conditions to avoid

Incompatible products. Excess heat.

### 10.5. Incompatible materials

None known.

### 10.6. Hazardous decomposition products

None under normal use conditions.

## Section 11: Toxicological information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008



# SAFETY DATA SHEET

50% v/v Ethylene Glycol Solution

Revision Date 23-Jul-2024

## Product Information

### (a) acute toxicity;

Oral

Based on ATE data, the classification criteria are not met  
ATE = 3052 mg/kg

Dermal

Based on ATE data, the classification criteria are not met

Inhalation

Based on ATE data, the classification criteria are not met

## Toxicology data for the components

| Component       | LD50 Oral                                                       | LD50 Dermal                                                                            | LC50 Inhalation             |
|-----------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------|
| Ethylene glycol | 7712 mg/kg ( Rat )<br>Human LD50 estimate:<br>1,400–1,600 mg/kg | LD50 = 9530 µL/kg ( Rabbit )<br>LD50 = 10600 mg/kg ( Rat )<br>LD50 > 3500 mg/kg (mice) | LC50 > 2.5 mg/L ( Rat ) 6 h |
| Water           | -                                                               | -                                                                                      | -                           |

### (b) skin corrosion/irritation;

Based on available data, the classification criteria are not met

### (c) serious eye damage/irritation;

Based on available data, the classification criteria are not met

### (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

### (f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

### (g) reproductive toxicity;

Based on available data, the classification criteria are not met

### (h) STOT-single exposure;

Based on available data, the classification criteria are not met

### (i) STOT-repeated exposure;

Based on available data, the classification criteria are not met

Target Organs

No information available.

### (j) aspiration hazard;

Based on available data, the classification criteria are not met

Symptoms / effects, both acute and delayed  
No information available.

## 11.2. Information on other hazards

### Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## Section 12: Ecological information

### 12.1. Toxicity

# SAFETY DATA SHEET

50% v/v Ethylene Glycol Solution

Revision Date 23-Jul-2024

## Ecotoxicity effects

| Component       | Freshwater Fish                                                                                                                                                                                                                                                                                                                                               | Water Flea                                 | Freshwater Algae                                                  |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------|
| Ethylene glycol | LC50: = 41000 mg/L, 96h<br>(Oncorhynchus mykiss)<br>LC50: = 27540 mg/L, 96h static<br>(Lepomis macrochirus)<br>LC50: 14 - 18 mL/L, 96h static<br>(Oncorhynchus mykiss)<br>LC50: = 40761 mg/L, 96h static<br>(Oncorhynchus mykiss)<br>LC50: 40000 - 60000 mg/L, 96h<br>static (Pimephales promelas)<br>LC50: = 16000 mg/L, 96h static<br>(Poecilia reticulata) | EC50: = 46300 mg/L, 48h<br>(Daphnia magna) | EC50: 6500 - 13000 mg/L, 96h<br>(Pseudokirchneriella subcapitata) |

## 12.2. Persistence and degradability

### Persistence

Soluble in water, Persistence is unlikely, based on information available.

## 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component       | log Pow | Bioconcentration factor (BCF) |
|-----------------|---------|-------------------------------|
| Ethylene glycol | -1.36   | No data available             |

## 12.4. Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

## 12.5. Results of PBT and vPvB assessment

No data available for assessment.

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13: Disposal considerations

## 13.1. Waste treatment methods

### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

### Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

### Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and

# SAFETY DATA SHEET

50% v/v Ethylene Glycol Solution

Revision Date 23-Jul-2024

regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## Section 14: Transport information

**IMDG/IMO** Not regulated

**14.1. UN number**  
**14.2. UN proper shipping name**  
**14.3. Transport hazard class(es)**  
**14.4. Packing group**

**ADR** Not regulated

**14.1. UN number**  
**14.2. UN proper shipping name**  
**14.3. Transport hazard class(es)**  
**14.4. Packing group**

**IATA** Not regulated

**14.1. UN number**  
**14.2. UN proper shipping name**  
**14.3. Transport hazard class(es)**  
**14.4. Packing group**

**14.5. Environmental hazards** No hazards identified

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## Section 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component       | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Ethylene glycol | 107-21-1  | 203-473-3 | -      | -   | X     | X    | KE-13169 | X    | X    |
| Water           | 7732-18-5 | 231-791-2 | -      | -   | X     | X    | KE-35400 | X    | -    |

| Component       | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Ethylene glycol | 107-21-1  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Water           | 7732-18-5 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**Authorisation/Restrictions according to EU REACH** Not applicable

| Component | CAS No | REACH (1907/2006) - | REACH (1907/2006) - | REACH Regulation (EC |
|-----------|--------|---------------------|---------------------|----------------------|
|-----------|--------|---------------------|---------------------|----------------------|

FSUTS0607

# SAFETY DATA SHEET

50% v/v Ethylene Glycol Solution

Revision Date 23-Jul-2024

|                 |           | Annex XIV - Substances Subject to Authorization | Annex XVII - Restrictions on Certain Dangerous Substances | 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------|-----------|-------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------|
| Ethylene glycol | 107-21-1  | -                                               | -                                                         | -                                                                                |
| Water           | 7732-18-5 | -                                               | -                                                         | -                                                                                |

## Seveso III Directive (2012/18/EC)

| Component       | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Ethylene glycol | 107-21-1  | Not applicable                                                                            | Not applicable                                                                           |
| Water           | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 1 (self classification)

| Component       | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------------|---------------------------------------|-------------------------|
| Ethylene glycol | WGK1                                  |                         |

| Component       | France - INRS (Tables of occupational diseases)      |
|-----------------|------------------------------------------------------|
| Ethylene glycol | Tableaux des maladies professionnelles (TMP) - RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                               | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Ethylene glycol<br>107-21-1 ( 50 - 60 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## Section 16: Other information

# SAFETY DATA SHEET

50% v/v Ethylene Glycol Solution

Revision Date 23-Jul-2024

## Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed

H373 - May cause damage to organs through prolonged or repeated exposure

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

**Creation Date** 22-Jul-2024

**Revision Date** 23-Jul-2024

**Revision Summary** Initial Release.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the

# **SAFETY DATA SHEET**

50% v/v Ethylene Glycol Solution

Revision Date 23-Jul-2024

---

date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**